



EV Intelligence

A Snowflake reference architecture on Microsoft Azure

Fleet electrification analysis for Contoso Logistics, built on Washington State's public EV registration data

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Questions the raw data can't answer

Contoso Logistics is electrifying its Washington fleet under a 200-mile minimum range policy. The fleet lead wants two things before the quarterly review:

- How many registered vehicles actually clear the bar
- How much of the data is usable enough to say so

The registration file has a defect that makes the obvious answer wrong.

Electric Range is zero for 8,155 records. Zero means the state never researched the range, not that the vehicle has none.

47.8%

of battery electric vehicles have no researched range

0%

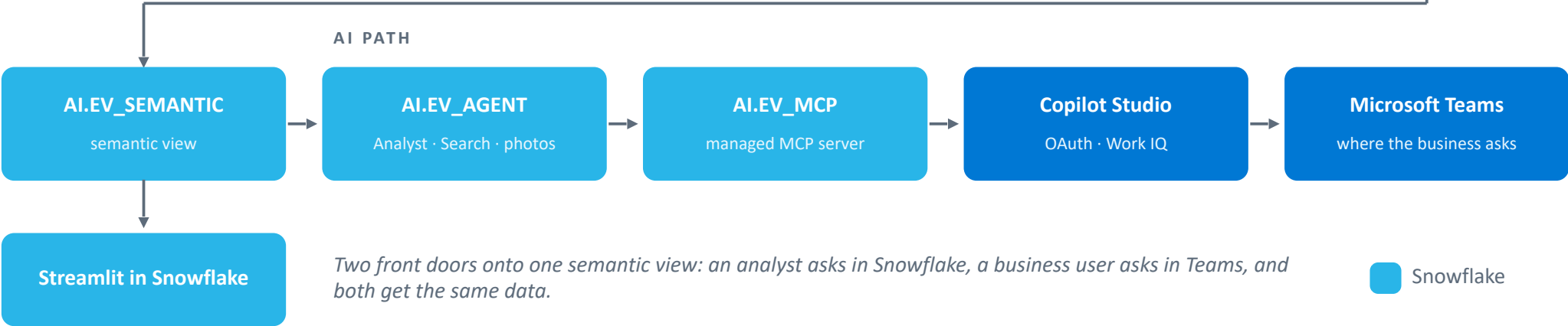
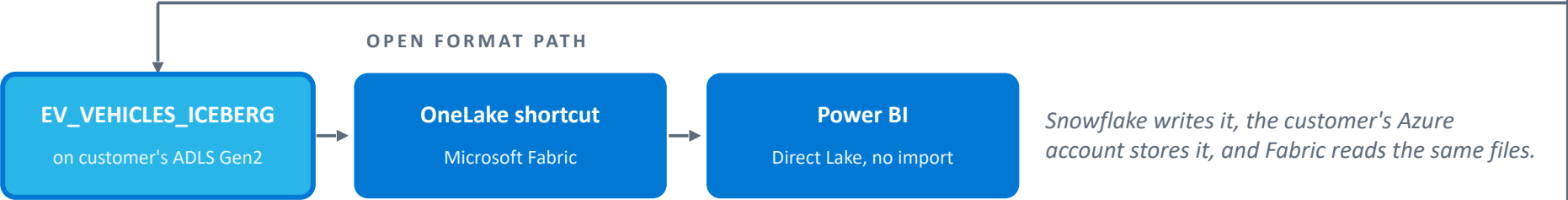
of plug-in hybrids are affected, so the gap is not random

-126 miles

The average impact of messy data on Tesla

One copy of the data, read by every engine

INGEST AND TRANSFORM



Cost: Event-driven ingestion, incremental processing, and auto-suspend minimize idle compute.

Governance: Least-privilege RBAC and VIN masking in Snowflake; Fabric access governed separately.

Architectural Decisions

CHOSE	OVER	BECAUSE
Snowpipe	Azure Data Factory	Event Grid already emits blob events.
Streams and tasks with dynamic tables	dbt or Snowpark alone	COPY cannot FLATTEN at ingest, so the task does the merge and dynamic tables handle the declarative layer above it.
Snowflake-managed Iceberg	Externally managed	Snowflake owns the writer and metadata, so DML and refreshes work normally. Files still land in the customer's storage.
OneLake shortcut	Catalog-linked database	Microsoft's documented pattern. The cost is that a shortcut points at a folder instead of a table.

Note: there are *many* ways Microsoft and Snowflake can operate together. Today’s demo takes a particular path but does not exhaust all possibilities.

Fabric + Snowflake: Each platform at its best

The joint case

- The primary data plane stays in Azure
- Snowflake is the engine; storage, BI and collaboration extend to Microsoft
- Running this solution drives Azure consumption and M365 usage at the same time as Snowflake spend
- Open by design: shared data, multiple engines, customer choice.

“Why not just use Fabric? Why not just use Snowflake?”

Why both? Extend what works instead of replacing it.

Snowflake and Fabric can share an open data layer while serving different users and workloads.

- **Preserve existing investments.** Add Fabric experiences to Snowflake data without forcing a platform migration.
- **Connect data to the tools people use.** Snowflake powers engineering, analytics, and AI; Fabric extends curated data into Power BI and Microsoft experiences.
- **Reduce the integration tax.** A shared Iceberg layer means fewer serving copies, fewer synchronization pipelines, and less reconciliation.

The architecture has to work for everyone in the room

Data Engineer

Does it run without babysitting?

Automated and observable. New files trigger ingestion automatically; pipe and refresh history show exactly what ran; the demo proves the path with a second file.

Analytics Leader

Can I trust the numbers?

Same curated data, aligned definitions. Power BI and Copilot start from the same curated data product, with shared tests and documented metric definitions.

Architect or CTO

What am I locking into?

The data is portable; platform logic is not. Shared Iceberg tables remain in customer-managed ADLS, while policies, pipelines, and semantic models would require rework.

Microsoft Account Team

Why would Microsoft co-sell it?

It expands the Microsoft footprint around Snowflake. Azure handles storage and events; Fabric and Power BI consume the shared layer; Copilot stays the “UI for AI”.

The business user stays in Teams & Copilot; Snowflake delivers the governed answer.

From working demo to repeatable co-sell motion

Package It

Public-data demo, GitHub quickstart, architecture diagram, demo script, and documented tenant IAM steps.

Target It

Snowflake-on-Azure customers expanding Fabric or Power BI or just trying to eliminate duplicated data pipelines.

Activate it

Run a joint discovery workshop, adapt the data and business questions, and leave with an owner, workload, and proof-of-value plan.

Let's see it in action...